Midterm Exam

Please do NOT write on the back side of the paper as they will not be scanned onto Gradescope and not considered when grading.

Name: _____

1. Let $X_1, \ldots, X_n$ be independent random variables with common density

$$f(x) = \theta x^{\theta-1},\ 0 < x < 1$$

with $\theta > 0$ unknown.

(a) Define $\overline{Y_n} = (Y_1 + \cdots + Y_n)/n$ based on $Y_i = -\log(X_i)$, $i = 1, \ldots, n$. Show that

$$\widehat{\theta}_{MLE} = 1/\overline{Y_n}$$

is the maximum likelihood estimator of θ.

(b) Show that $Y_i = -\log(X_i)$ has density

$$f(y) = \theta \exp(-\theta y),\ y > 0.$$

You may use for the remainder of the question that

$$\mathrm{E}[Y_i] = 1/\theta \text{ and } \mathrm{Var}(Y_i) = 1/\theta^2.$$

(c) Show that $\widehat{\theta}_{MLE}$ is consistent.

(d) Compute the asymptotic variance $1/(nI(\theta))$ of $\widehat{\theta}_{MLE}$.

2. Let $X_1, \ldots, X_n$ be independent random variables with a common Poisson(μ) distribution. Recall

$$P(X = x|\mu) = \mu^x \exp(-\mu)/x! \qquad \text{for } x = 0, 1, 2, \cdots$$

$E(X) = \mu$ and $\mathrm{Var}(X) = \mu$. We are interested in the parameter

$$\theta := P(X = 0).$$

We propose the plug-in estimator $\widehat{\theta} := P(X = 0|\widehat{\mu})$ based on the maximum likelihood estimate $\widehat{\mu} = \overline{X}$ of μ.

(a) Show that $\widehat{\theta}$ is asymptotically unbiased.

(b) Is $\widehat{\theta}$ consistent? Explain.

(c) Find the asymptotic variance of $\widehat{\theta}$.

(d) Compare the asymptotic variance in part (c) with the variance of the naive estimator $\widetilde{\theta} = \sum_{i=1}^n 1\{X_i = 0\}/n$.

3. Let $X_1, \ldots, X_n$ be independent random variables with common density

$$f(x|\theta) = \begin{cases} 3e^{-3(x-\theta)} & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

for some parameter $\theta \in \mathbb{R}$.

(a) Find the maximum likelihood estimator of θ.